

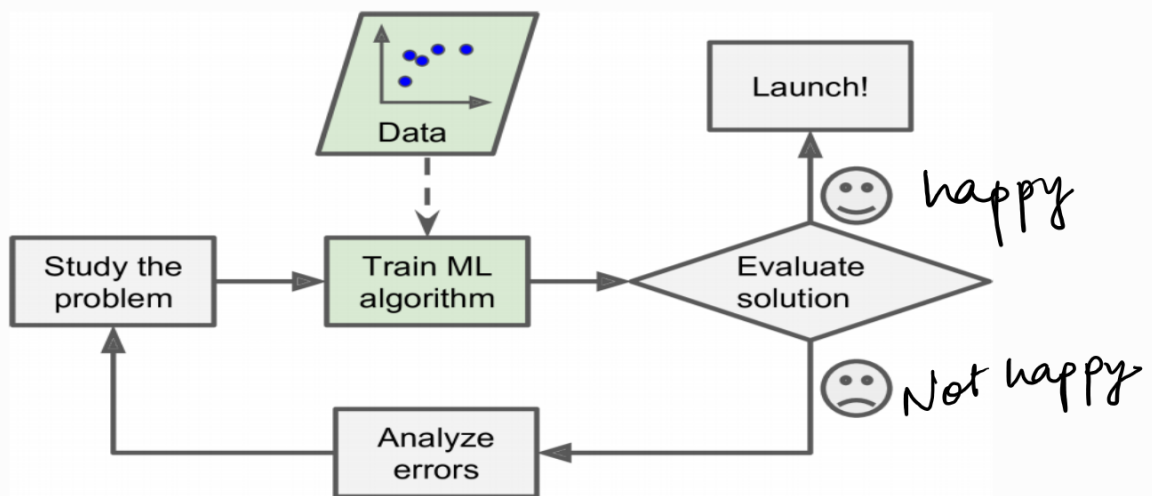
Machine learning

Def:- ML is the science of programming computers so they can learn from data.

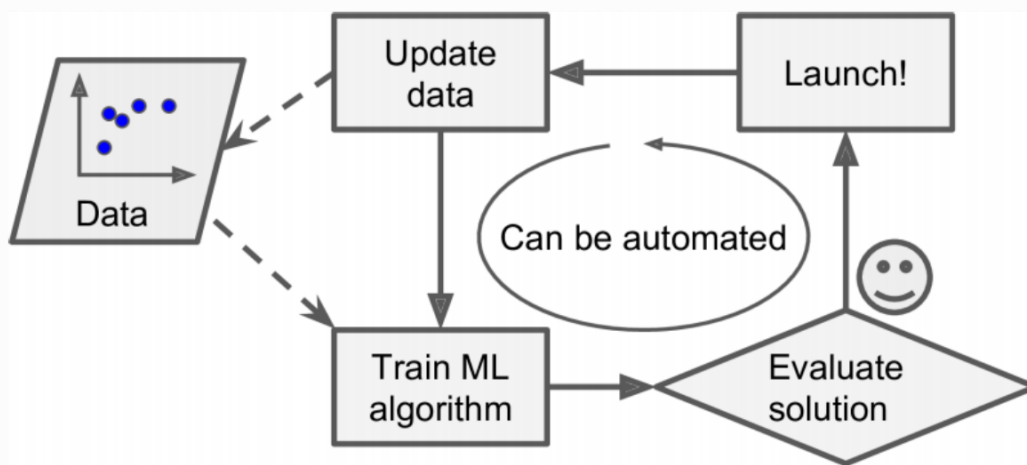
or

ML is the field of study that gives computers the ability to learn without being explicitly programmed

* the examples that the system uses to learn are called training set.

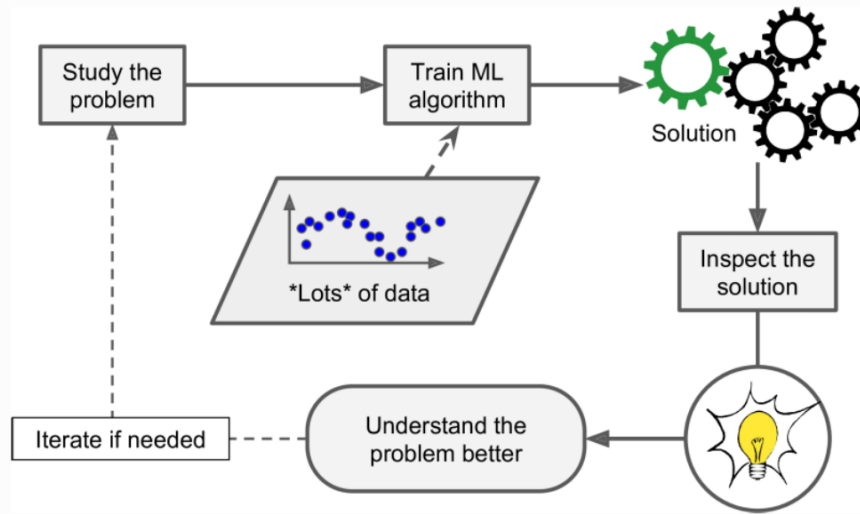


→ the ML approach



→ Automatically adapting to change

- * Applying ML techniques to dig into large amount of data can help discover patterns that were not immediately apparent. This is called data mining.



→ ML can help humans learn

To summarize, Machine Learning is great for:

- **Problems for which existing solutions require a lot of hand-tuning or long lists of rules:** one Machine Learning algorithm can often simplify code and perform better.
- **Complex problems for which there is no good solution at all using a traditional approach:** the best Machine Learning techniques can find a solution.
- **Fluctuating environments:** a Machine Learning system can adapt to new data.
- **Getting insights about complex problems and large amounts of data.**

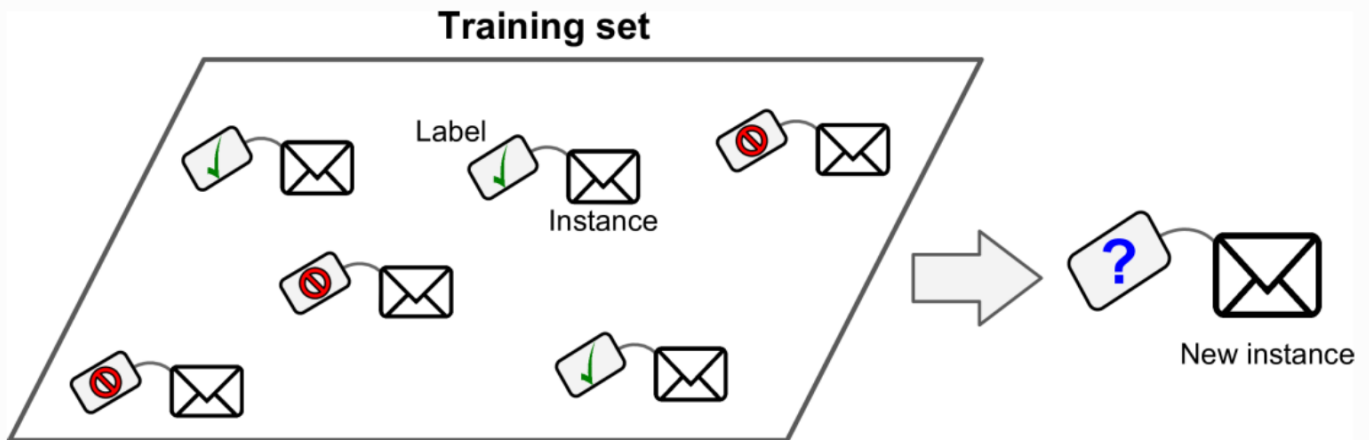
→ Types of ML systems

↳ Supervised, Unsupervised, semisupervised and Reinforcement learning

↳ Online versus batch learning

↳ Instance based versus model based learning

→ Supervised learning :- In this the training set you feed to the algorithms includes the desired solⁿ called labels.

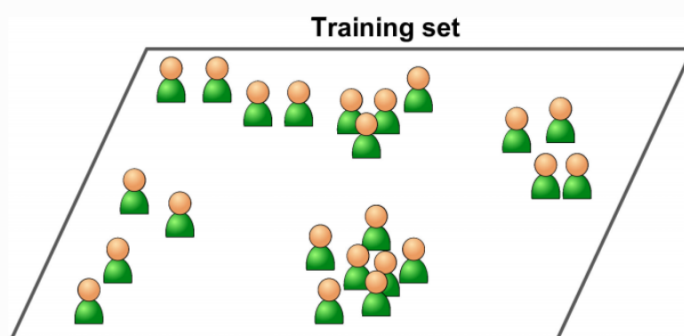


→ classification :- eg: spam filter
→ regression :- eg:- price of car, price of stock.

→ algorithms of supervised learning are :-

- k-Nearest Neighbors
- Linear Regression
- Logistic Regression
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural networks

→ Unsupervised Learning :- In this the training data is unlabeled. the system tries to learn without a teacher.



→ unlabeled data